Cross-disciplinary applications of the Toulbar2 solver

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Abstract / Introduction

This section introduces the[1] context of your research. Keep it brief. Use bullet points for readability:

- First important point of context.
- The main problem you are addressing.
- The goal of this state-of-the-art review.

Mathematical / Theoretical Background

Define the core concepts here. For example, a formal definition:

Let X be a topological space and $f : X \to \mathbb{R}$ a function. We define

You can include an image to illustrate the concept:

[Place a schematic or diagram here]

State of the Art: Main Approaches

This is the core of your poster. Compare the existing methods.

1. Classical Methods (e.g., 1990s-2010s)

- **Method A:** Brief description of pros and cons.
- **Method B:** Mathematical formulation of Method B.

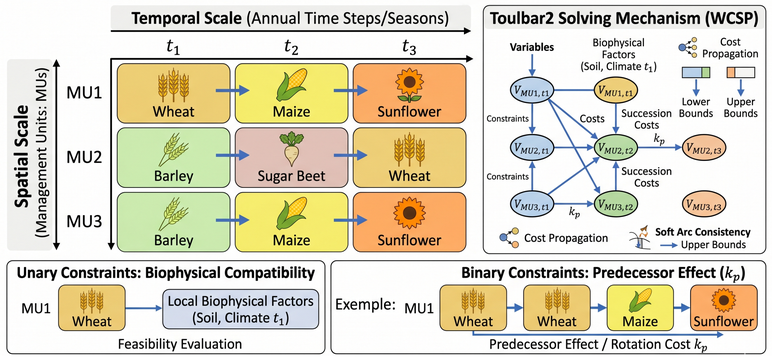
2. Modern Approaches (e.g., Deep Learning / Modern Heuristics)

- **Method C:** How it outperforms classical methods.
- Limitations still present today.

Recherche Opérationnelle & Planification

Le formalisme $WCSP$ performe dans la modélisation d'environnements complexes en combinant des règles strictes (contraintes dures) et l'optimisation de critères de qualité (contraintes souples).

Figure 1: Spatio-temporal Agricultural Modeling



Identified Limitations & Open Problems

Based on your review, what is missing in the current literature?

- Scalability issues for large datasets.
- Lack of theoretical bounds for algorithm Y.

Conclusion

Brief summary of the main takeaway from your literature review. What is the logical next step for your introductory research project?

References

[1] Pierre Montalbano, Simon Givry, and George Katsirelos. "Multiple-choice Knapsack Constraint in Graphical Models". en. In: *Integration of Constraint Programming, Artificial Intelligence, and Operations Research*. Ed. by Pierre Schaus. Cham: Springer International Publishing, 2022, pp. 282–299. ISBN: 978-3-031-08011-1. DOI: 10.1007/978-3-031-08011-1_19.

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